accepted and programmed into the skittish on-board computer. During that same week the JPL engineers designed an innovative command frequency control procedure to make sure that essential orders would be understood by the damaged backup receiver.

The engineers were now able to recommunicate, at least in a rudimentary way, with the spacecraft. Unfortunately the backup receiver now turned giddy, becoming extremely sensitive to the stray heat dumped when various components of the spacecraft powered up or down. In the following months the JPL engineers devised and conducted tests that let them thoroughly understand the thermal implications of most spacecraft operational modes: What would prevent and what would permit receipt of commands from Earth?

With this information, the backup receiver problem was entirely circumvented. It subsequently acquired all the commands sent from Earth on how to gather data in the Jupiter, Saturn, Uranus, and Neptune systems. The engineers had saved the mission. (To be on the safe side, during most of *Voyager 2*'s subsequent flight a nominal data-taking sequence for the next planet to be encountered was always sitting in the on-board computers—should the spacecraft again become deaf to entreaties from home.)

Another heart-wrenching failure occurred just after *Voyager 2* emerged from behind Saturn (as seen from the Earth) in August 1981. The scan platform had been moving feverishly—pointing here and there among the rings, moons, and the planet itself during the all-too-brief moments of close approach. Suddenly, the platform jammed. A stuck scan platform is a maddening predicament: knowing that the spacecraft is flying past wonders that have never been witnessed, that we will not see again for years or decades, and the incurious spacecraft staring fixedly off into space, ignoring everything.

The scan platform is driven by actuators containing gear trains. So first the JPL engineers ran an identical copy of a flight actuator in a simulated mission. This actuator failed after 348 turns; the actuator on the spacecraft had failed after 352 turns. The problem turned out